Scaling LLM-Enhanced Linear Autoencoders to Industrial-Size Catalogs

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Abstract

LLM-enhanced linear autoencoders (L^3AE) incorporate semantic item representations from large language models into collaborative filtering and demonstrate significant gains on long-tail items. However, L^3AE requires dense $n \times n$ matrices, which makes it impractical for large catalogs — exactly where semantic enrichment would help most. We propose scalable formulations that factor the semantic similarity matrix into a compact low-rank form and compute all required precision-matrix products without ever building any dense $n \times n$ matrix. Our method preserves the algebraic structure of L^3AE while replacing its dense collaborative precision with SANSA's sparse approximate inverse, enabling efficient scaling to industrial-size catalogs. Experiments on several public datasets show consistent gains over multiple baselines, and these gains tend to grow in colder, sparser catalogs — a pattern consistent with semantic enrichment helping most where collaborative signals are weakest. Code and configs are available at https://github.com/ViV99/scal3ae.

CCS Concepts

• **Information systems** → **Collaborative filtering**; **Recommender systems**.

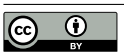
Keywords

collaborative filtering, linear autoencoders, large language models, scalability, long-tail recommendation

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1 Introduction

Linear autoencoders (LAEs) are a strong class of models for collaborative filtering (CF). Starting with EASE [8], they predict user preferences by learning an item-to-item weight matrix in closed form. Sparse variants such as SANSA [7] scale LAEs to industrial-size catalogs by exploiting the sparsity of item co-occurrence statistics.



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Recently, L^3AE [4] incorporated LLM-derived semantic embeddings into the LAE framework through a knowledge-distillation regularizer, yielding large accuracy gains on long-tail items with few interactions. However, L^3AE relies on dense $n \times n$ matrices with $O(n^2)$ memory — prohibitive for the large, sparse catalogs where semantic enrichment is most valuable.

To address this problem, we propose ScaL^{3}AE (**Sca**lable **L^{3}AE**), which utilizes two key ideas: (1) the semantic similarity matrix of L^3AE admits a compact factorization into a diagonal part and a low-rank part whose dimension equals the embedding size; and (2) all remaining dependencies on the dense precision matrix can be evaluated through sparse approximate inverse computations borrowed from SANSA.

Our contributions:

- **ScaL^{3}AE**: an industrial-scale formulation of L^3AE that combines Woodbury factorization of the semantic similarity matrix with SANSA's sparse approximate inverse, preserving the algebraic structure of L^3AE without materializing any dense $n \times n$ matrix.
- **Large-scale evaluation** across 4 Amazon product domains at three k-core filtering thresholds ($k \in \{10, 5, 3\}$), with consistent gains over SANSA that tend to grow in colder, sparser catalogs.
- **Open release**: code for the model and a pipeline that reproduces all experiments are available at https://github.com/ViV99/scal3ae.

2 Background

Problem setting. Let $\mathbf{X} \in \{0,1\}^{m \times n}$ be a binary user–item interaction matrix with m users and n items. LAEs learn an item-to-item weight matrix $\mathbf{B} \in \mathbb{R}^{n \times n}$ so that for user u, the predicted scores are $\hat{\mathbf{y}}_u = \mathbf{x}_u \mathbf{B}$, where $\mathbf{x}_u$ is the u-th row of $\mathbf{X}$.

EASE^R [8]. For regularization strength $\lambda > 0$, EASE minimizes $\|\mathbf{X} - \mathbf{X}\mathbf{B}\|_F^2 + \lambda \|\mathbf{B}\|_F^2$ with the constraint $\mathrm{diag}(\mathbf{B}) = \mathbf{0}$. The closed-form solution is:

$$\mathbf{B}_{\mathrm{EASE}} = \mathbf{I} - \mathbf{P} \cdot \mathrm{Diag}(\boldsymbol{\mu}), \qquad \mathbf{P} = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1}, \tag{1}$$

where $\mathbf{P}$ is the collaborative precision matrix and $\boldsymbol{\mu} \in \mathbb{R}^n$ is the vector with entries $\mu_j = 1/P_{jj}$ (with P_{jj} the j-th diagonal entry of $\mathbf{P}$) that enforces $\mathrm{diag}(\mathbf{B}) = \mathbf{0}$. EASE needs $O(n^2)$ memory and $O(n^3)$ time.

SANSA [7]. SANSA approximates EASE through a sparse approximate factorization of the Gram matrix $\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}$, producing a sparse weight matrix $\mathbf{B}_{\mathrm{SANSA}}$ of target density $\rho \in (0, 1]$ (so $\mathrm{nnz}(\mathbf{B}_{\mathrm{SANSA}}) = \rho n^2$, where $\mathrm{nnz}(\cdot)$ counts nonzero entries). Memory drops to $O(\rho n^2)$, which allows scaling to millions of items.

ELSA [9]. ELSA scales LAEs by constraining the item–item weight matrix to a low-rank factorization, achieving sub-quadratic memory like SANSA but through low rank rather than sparsity. It is the closest structural relative of ScaL^{3}AE, which likewise lands on a low-rank-plus-sparse form; the difference is that ELSA's low rank is a purely collaborative compression of the interaction matrix, whereas ScaL^{3}AE's low-rank factor originates from the LLM semantic matrix $\mathbf{S}$ — so ScaL^{3}AE adds semantic signal that ELSA does not model.

L^3AE [4]. L^{3}AE adds semantic knowledge from LLM embeddings $\mathbf{F} \in \mathbb{R}^{d\times n}$ (with embedding dimension $d \ll n$, one column per item) to EASE. In Phase 1, it builds a semantic item correlation matrix by applying the EASE closed form to $\mathbf{F}$:

$$\mathbf{S} = \mathbf{I} - \mathbf{P}_F \cdot \mathrm{Diag}(\boldsymbol{\gamma}_F), \qquad \mathbf{P}_F = (\mathbf{F}^\top\mathbf{F} + \lambda_F\mathbf{I})^{-1}, \tag{2}$$

where $\mathbf{P}_F$ is the semantic precision matrix, and $\boldsymbol{\gamma}_F \in \mathbb{R}^n$ has entries $\gamma_{F,j} = 1/(P_F)_{jj}$ so that $\mathrm{diag}(\mathbf{S}) = \mathbf{0}$. In Phase 2, the weight matrix $\mathbf{B}$ is learned from a knowledge-distillation (KD) objective that balances a Tikhonov-regularized fit to the interaction matrix (weight λ_X, analogous to EASE's λ) against regularization toward the semantic similarity $\mathbf{S}$ (KD strength λ_{KD}). The closed-form solution is

$$\mathbf{B}_{L^3AE} = \mathbf{I} + \lambda_{\mathrm{KD}}\,\mathbf{P}_{\mathrm{KD}}\mathbf{S} - \mathbf{P}_{\mathrm{KD}}\,\mathrm{Diag}(\boldsymbol{\mu}), \quad \mathbf{P}_{\mathrm{KD}} = (\mathbf{X}^\top\mathbf{X} + \lambda'\mathbf{I})^{-1}, \tag{3}$$

where $\lambda' = \lambda_{\mathrm{KD}} + \lambda_X$ and $\boldsymbol{\mu} \in \mathbb{R}^n$ is redefined relative to Eq. (1) as $\mu_j = (1+\lambda_{\mathrm{KD}}(\mathbf{P}_{\mathrm{KD}}\mathbf{S})_{jj})/(P_{\mathrm{KD}})_{jj}$, again enforcing $\mathrm{diag}(\mathbf{B}_{L^3AE}) = \mathbf{0}$. The term $\lambda_{\mathrm{KD}}\,\mathbf{P}_{\mathrm{KD}}\mathbf{S}$ pulls $\mathbf{B}$ toward semantic item similarities and helps most for long-tail items. Both $\mathbf{P}_{\mathrm{KD}}$ and $\mathbf{S}$ are dense $n \times n$, so L^{3}AE becomes impractical for large catalogs.

3 Method: ScaL^{3}AE

The bottleneck in L^{3}AE is the product $\mathbf{P}_{\mathrm{KD}}\mathbf{S}$, where both matrices are dense $n\times n$. ScaL^{3}AE removes this bottleneck in two steps: (1) we factor $\mathbf{S}$ into diagonal plus low-rank form using the Woodbury identity (§3.1), and (2) we compute all products with $\mathbf{P}_{\mathrm{KD}}$ through a sparse approximate inverse (§3.2), which gives an efficient inference formula (§3.3).

3.1 Woodbury Factorization of S

Since the embedding dimension $d \ll n$, we can apply the Woodbury identity to $\mathbf{P}_F = (\mathbf{F}^\top\mathbf{F} + \lambda_F\mathbf{I}_n)^{-1}$:

$$\mathbf{P}_F = \frac{1}{\lambda_F}\left(\mathbf{I}_n - \mathbf{F}^\top\mathbf{G}_F\mathbf{F}\right), \quad \mathbf{G}_F = (\lambda_F\mathbf{I}_d + \mathbf{F}\mathbf{F}^\top)^{-1}, \tag{4}$$

where $\mathbf{G}_F$ needs only a $d \times d$ inversion. Plugging this into Eq. (2) gives a factored form of $\mathbf{S}$:

$$\mathbf{S} = \mathrm{Diag}(\mathbf{d}_S) + \mathbf{F}^\top\mathbf{C}, \tag{5}$$

where $\mathbf{d}_S \in \mathbb{R}^n$ is the diagonal part (with j-th entry $d_{S,j}$) and $\mathbf{C} \in \mathbb{R}^{d\times n}$ is the low-rank factor:

$$d_{S,j} = 1 - \frac{\gamma_{F,j}}{\lambda_F}, \qquad \mathbf{C} = \frac{1}{\lambda_F}\mathbf{G}_F\mathbf{F}\,\mathrm{Diag}(\boldsymbol{\gamma}_F). \tag{6}$$

The diagonal entries of $\mathbf{P}_F$ needed for $\boldsymbol{\gamma}_F$ (Eq. (2)) can be read off Eq. (4): $(P_F)_{jj} = \frac{1}{\lambda_F}(1 - \mathbf{f}_j^\top\mathbf{G}_F\mathbf{f}_j)$, where $\mathbf{f}_j$ is the j-th column of $\mathbf{F}$. Crucially, the $n \times n$ matrix $\mathbf{S}$ is never materialized; we only store $\mathbf{d}_S$ (n values) and $\mathbf{C}$ ($d \times n$).

3.2 Computing Precision-Matrix Products

With $\mathbf{S}$ factored, the semantic term from Eq. (3) becomes:

$$\mathbf{P}_{\mathrm{KD}}\mathbf{S} = \mathbf{P}_{\mathrm{KD}}\,\mathrm{Diag}(\mathbf{d}_S) + \underbrace{\mathbf{P}_{\mathrm{KD}}\mathbf{F}^\top}_{\boldsymbol{\Psi}\in\mathbb{R}^{n\times d}} \cdot \mathbf{C}. \tag{7}$$

We therefore need only two quantities from the dense $\mathbf{P}_{\mathrm{KD}}$: the thin matrix $\boldsymbol{\Psi} = \mathbf{P}_{\mathrm{KD}}\mathbf{F}^\top$ and the vector $\mathrm{diag}(\mathbf{P}_{\mathrm{KD}}) \in \mathbb{R}^n$. Instead of building $\mathbf{P}_{\mathrm{KD}}$, we obtain these from a sparse factorization produced by SANSA.

Let $\mathbf{P}_{\mathrm{KD}} \approx \mathbf{D}_{pc}^{-1}\,\mathbf{K}^\top\,\hat{\mathbf{D}}^{-1}\,\mathbf{K}\,\mathbf{D}_{pc}^{-1}$, SANSA's preconditioned LDL-type approximate inverse, where $\mathbf{K} \in \mathbb{R}^{n\times n}$ is sparse, $\mathbf{D}_{pc}, \hat{\mathbf{D}} \in \mathbb{R}^{n\times n}$ are diagonal, and $(D_{pc})_{jj} = \sqrt{(\mathbf{X}^\top\mathbf{X})_{jj} + \lambda'}$ is a diagonal preconditioner we add for numerical stability (any sparse factorization that yields such a chain of factors for $\mathbf{P}_{\mathrm{KD}}$ works equally well). Right-multiplying this chain by $\mathbf{F}^\top$ gives

$$\boldsymbol{\Psi} = \mathbf{D}_{pc}^{-1}\,\mathbf{K}^\top\,\hat{\mathbf{D}}^{-1}\,\mathbf{K}\,\mathbf{D}_{pc}^{-1}\,\mathbf{F}^\top, \tag{8}$$

evaluated right-to-left: each step is a sparse × dense product, so the total cost is $O(\mathrm{nnz}(\mathbf{K}) \cdot d)$. The diagonal of the precision matrix is obtained from the same factors as $(P_{\mathrm{KD}})_{jj} = (D_{pc})_{jj}^{-2}\sum_i K_{ij}^2/\hat{D}_{ii}$, in $O(\mathrm{nnz}(\mathbf{K}))$ time. The dense $n\times n$ matrix $\mathbf{P}_{\mathrm{KD}}$ is never materialized.

3.3 Inference

Let $p_j = (P_{\mathrm{KD}})_{jj}$ denote the j-th diagonal entry of $\mathbf{P}_{\mathrm{KD}}$, obtainable from the SANSA factors in $O(\mathrm{nnz}(\mathbf{K}))$ time (§3.2). For a user with interaction vector $\mathbf{x}$, the predicted scores are $\hat{\mathbf{y}} = \mathbf{x}\,\mathbf{B}_{L^3AE}$. We rewrite this in a form that uses only $\boldsymbol{\Psi}$, $\mathbf{C}$, $\mathbf{d}_S$, $\mathbf{p}$, and sparse products against $\mathbf{B}_{\mathrm{SANSA}}$ — never touching any dense $n\times n$ matrix. Expanding Eq. (3) and substituting the factored $\mathbf{S}$ from Eq. (5) gives

$$\mathbf{x}\,\mathbf{B}_{L^3AE} = \mathbf{x} + (\mathbf{x}\,\mathbf{P}_{\mathrm{KD}}) \odot \left(\lambda_{\mathrm{KD}}\mathbf{d}_S - \boldsymbol{\mu}\right) + \lambda_{\mathrm{KD}}(\mathbf{x}\,\boldsymbol{\Psi})\,\mathbf{C}. \tag{9}$$

SANSA approximates the EASE form of $\mathbf{P}_{\mathrm{KD}}$, namely $\mathbf{B}_{\mathrm{SANSA}} \approx \mathbf{I} - \mathbf{P}_{\mathrm{KD}}\,\mathrm{Diag}(1/\mathbf{p})$, so $\mathbf{x}\,\mathbf{P}_{\mathrm{KD}} = (\mathbf{x} - \mathbf{x}\,\mathbf{B}_{\mathrm{SANSA}}) \odot \mathbf{p}$; plugging this in and regrouping (using $p_j\mu_j = 1 + \lambda_{\mathrm{KD}}v_j$ with $v_j = p_j d_{S,j} + (\boldsymbol{\Psi}\mathbf{C})_{jj}$ from Eq. (7)) yields

$$\hat{\mathbf{y}} = \underbrace{\mathbf{x}\,\mathbf{B}_{\mathrm{SANSA}}}_{\text{collaborative}} + \underbrace{(\mathbf{x} - \mathbf{x}\,\mathbf{B}_{\mathrm{SANSA}}) \odot \mathbf{w}}_{\text{residual correction}} + \underbrace{\lambda_{\mathrm{KD}}\,(\mathbf{x}\,\boldsymbol{\Psi})\,\mathbf{C}}_{\text{semantic (rank } d)}, \tag{10}$$

with the precomputed correction vector

$$w_j = -\lambda_{\mathrm{KD}}\sum_{k=1}^{d}\Psi_{jk}\,C_{kj}, \tag{11}$$

which absorbs the zero-diagonal constraint $(B_{L^3AE})_{jj} = 0$ carried by $\boldsymbol{\mu}$ in Eq. (3). The first term is the standard SANSA prediction (sparse × sparse). The second term adjusts scores using the SANSA residual — items that SANSA could not explain well receive a stronger semantic correction. The third term is the low-rank semantic part: it projects the user into the d-dimensional embedding space and maps back to item scores.

3.4 Complexity

Table 1 summarizes the asymptotic cost of each method. ScaL^{3}AE inherits SANSA's sub-quadratic memory while paying only a small $O(dn)$ overhead for the low-rank factors $\boldsymbol{\Psi}$ and $\mathbf{C}$; compared to

Table 1: Asymptotic cost of EASE, SANSA, L^{3}AE, and ScaL^{3}AE. Here n is the number of items, k the user history length, d the embedding dimension, and $\rho \in (0, 1]$ the target density of sparse factors. "Info" indicates whether the model uses collaborative (C) signal, semantic (S) signal, or both.

Method	Info	Memory	Train	Infer / user
EASE	C	$O(n^2)$	$O(n^3)$	$O(kn)$
SANSA	C	$O(\rho n^2)$	$O(\rho n^2)$	$O(k\rho n)$
L^{3}AE	C+S	$O(n^2)$	$O(n^3)$	$O(kn)$
ScaL^{3}AE	C+S	$O(\rho n^2 + dn)$	$O(\rho n^2 d)$	$O(k\rho n + dn)$

EASE and L^{3}AE, memory drops from $O(n^2)$ to $O(\rho n^2 + dn)$, training from $O(n^3)$ to $O(\rho n^2 d)$, and per-user inference from $O(kn)$ to $O(k\rho n + dn)$. We report dominant asymptotic costs and omit lower-order terms such as the $d \times d$ inversion in ScaL^{3}AE's Woodbury identity.

4 Experiments

4.1 Experimental Setup

We organize the evaluation around four research questions:

- **RQ1** Does ScaL^{3}AE scale to catalog sizes where the dense L^{3}AE formulation is no longer tractable?
- **RQ2** Does ScaL^{3}AE match L^{3}AE quality on small catalogs and offer a favorable quality–scalability trade-off across all scales?
- **RQ3** How do semantic gains change as the catalog gets colder (more cold items)?
- **RQ4** How much does the residual-correction term contribute to ScaL^{3}AE's performance?

Datasets. We evaluate on four product domains from Amazon Reviews 2023 [3]: *Games*, *Toys*, *Books*, and *Electronics*. Each domain is preprocessed with k-core filtering at three thresholds $k \in \{10, 5, 3\}$ (at each threshold we iteratively remove users and items with fewer than k interactions until all remaining nodes satisfy the bound, following [4]), producing a wide range of catalog scales spanning three orders of magnitude — from Games-10 (~3K items, ~6K users) to Books-3 (~964K items, ~1.7M users) and Electronics-3 (~533K items, ~3.0M users). Interactions with ratings ≥ 4 are binarized, and each dataset is split chronologically into 80%/10%/10% train/validation/test sets.

Embeddings. We use BGE-Large [10] (d=1024) as the primary embedding model to encode item text descriptions, and also run experiments with Qwen3-Embedding-8B [11] (d=4096) to test robustness to the choice of embedding model.

Baselines. We compare with: **(i)** collaborative LAEs: EASE [8] (dense; small catalogs only), SANSA [7] (sparse), ELSA [9] (low-rank); **(ii)** graph-based: SGFCF [5], LightGCN [2]; **(iii)** LLM-enhanced neural: AlphaRec [6], VBPR [1]; **(iv)** dense semantic LAE: L^{3}AE [4] (dense; small catalogs only).

Metrics. We report Recall@20 and NDCG@20, computed over all unrated items, as mean ± standard deviation across bootstrap resamples of the test set; we focus discussion on gaps that are materially larger than this variability. For the head/tail breakdown

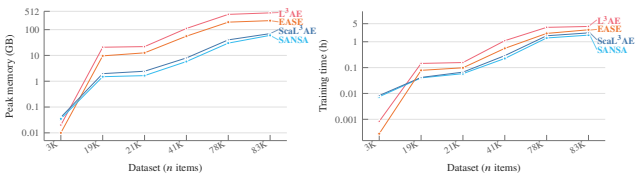


Figure 1: Peak memory (left) and training time (right) as a function of catalog size (n items, log scale). SANSA and ScaL^{3}AE are run at density $\rho = 0.01$.

we split items into *head* (top 20% by training popularity) and *tail* (remaining 80%), and bucket each test user by whether their first held-out item is in the head or tail set. Recall@20 is then computed on that user subset under the same full-catalog protocol, giving a simple user-level decomposition under chronological multi-target evaluation rather than a per-item attribution of every held-out interaction.

Hyperparameters. All methods are tuned per (dataset, embedding) by validation Recall@20; for SANSA and ScaL^{3}AE we select the largest target density ρ whose factorization fits into the 512 GB RAM available on our compute (per-dataset values in §4.2). The KD strength λ_{KD} interpolates between a purely collaborative model ($\lambda_{\text{KD}} \to 0$, recovering the SANSA solution) and a strongly semantic one; we select it per setting by validation Recall@20 and leave a systematic sensitivity analysis of λ_{KD} to future work.

4.2 Scalability (RQ1)

Figure 1 plots measured peak memory and training time across six datasets spanning 3K–83K items. EASE and L^{3}AE exhaust memory around $n \approx 10^5$ items and fail to produce a model on the larger catalogs, while SANSA and ScaL^{3}AE scale to catalogs with up to ~1M items. The gap between SANSA and ScaL^{3}AE stays within a thin $O(dn)$ band corresponding to the low-rank factors $\mathbf{\Psi}$ and $\mathbf{C}$.

At the extreme end — Electronics-3 (533 K items) and Books-3 (964 K items) — SANSA and ScaL^{3}AE remain tractable at ρ=10^{-4}, peaking at 24.5/35.0 GB and 146/169 GB respectively with training times $\leq$ 24.3 h; EASE and L^{3}AE are intractable at this scale.

4.3 Recommendation Quality (RQ2)

Tables 2 and 3 report Recall@20 and NDCG@20 on all four domains, under both BGE-Large and Qwen3-Embedding for the embedding-dependent methods. Reported ScaL^{3}AE numbers use the derivation-consistent default with the residual correction term in Eq. (10); we analyze the **w=0** variant in §4.5.

ScaL^{3}AE outperforms the collaborative baselines (EASE, SANSA) on every dataset and is the best per-column Recall@20 on 8/12 settings among BGE-based methods and 6/12 among Qwen-based ones (7/12 and 6/12 for NDCG@20); where L^{3}AE is tractable, ScaL^{3}AE matches or exceeds it, showing that the sparse approximation does not sacrifice quality. The remaining wins go to neural AlphaRec on Games and Electronics, a pattern consistent across BGE and Qwen that we examine next.

The Games/Electronics gap is driven almost entirely by head items: ScaL^{3}AE keeps head and tail recall near parity across the

Table 2: Recall@20 across 4 Amazon product domains. Per column, best among BGE-based methods in bold, best among Qwen-based methods underlined (non-semantic baselines count in both subsets). — indicates the method exceeded available memory (512 GB).

	Games			Toys			Books			Electronics		
Model	10	5	3	10	5	3	10	5	3	10	5	3
EASE	$.0339_{\pm.0054}$	$.0165_{\pm.0016}$	$.0142_{\pm.0010}$	$.0118_{\pm.0021}$	—	—	$.0280_{\pm.0012}$	—	—	$.0102_{\pm.0005}$	—	—
SANSA	$.0346_{\pm.0055}$	$.0130_{\pm.0014}$	$.0087_{\pm.0007}$	$.0105_{\pm.0019}$	$.0062_{\pm.0005}$	$.0038_{\pm.0002}$	$.0294_{\pm.0013}$	$.0169_{\pm.0005}$	$.0109_{\pm.0003}$	$.0090_{\pm.0005}$	$.0054_{\pm.0002}$	$.0041_{\pm.0001}$
ELSA	$.0367_{\pm.0059}$	$.0170_{\pm.0016}$	$.0138_{\pm.0010}$	$.0111_{\pm.0019}$	$.0071_{\pm.0006}$	$.0061_{\pm.0003}$	$.0266_{\pm.0012}$	$.0146_{\pm.0005}$	$.0106_{\pm.0003}$	$.0104_{\pm.0005}$	$.0105_{\pm.0003}$	$.0095_{\pm.0002}$
L^3AE (BGE)	$.0360_{\pm.0056}$	$.0190_{\pm.0017}$	$.0152_{\pm.0010}$	$.0143_{\pm.0023}$	—	—	$.0268_{\pm.0012}$	—	—	$.0105_{\pm.0005}$	—	—
L^3AE (Qwen)	$.0360_{\pm.0055}$	$.0191_{\pm.0017}$	$.0155_{\pm.0010}$	$.0159_{\pm.0024}$	—	—	$.0270_{\pm.0012}$	—	—	$.0107_{\pm.0005}$	—	—
SGFCF	$.0383_{\pm.0056}$	$.0196_{\pm.0017}$	$.0140_{\pm.0009}$	$.0125_{\pm.0020}$	$.0064_{\pm.0005}$	$.0048_{\pm.0003}$	$.0172_{\pm.0010}$	$.0069_{\pm.0003}$	$.0033_{\pm.0001}$	$.0092_{\pm.0005}$	$.0082_{\pm.0002}$	$.0084_{\pm.0002}$
LightGCN	$.0308_{\pm.0050}$	$.0186_{\pm.0017}$	$.0163_{\pm.0010}$	$.0078_{\pm.0014}$	$.0074_{\pm.0006}$	$.0062_{\pm.0003}$	$.0145_{\pm.0009}$	$.0095_{\pm.0004}$	$.0072_{\pm.0002}$	$.0080_{\pm.0004}$	$.0082_{\pm.0002}$	$.0085_{\pm.0002}$
VBPR (BGE)	$.0311_{\pm.0053}$	$.0135_{\pm.0014}$	$.0116_{\pm.0009}$	$.0096_{\pm.0017}$	$.0059_{\pm.0005}$	$.0041_{\pm.0003}$	$.0167_{\pm.0010}$	$.0096_{\pm.0004}$	$.0074_{\pm.0002}$	$.0072_{\pm.0004}$	$.0064_{\pm.0002}$	$.0064_{\pm.0002}$
VBPR (Qwen)	$.0308_{\pm.0052}$	$.0150_{\pm.0015}$	$.0112_{\pm.0009}$	$.0101_{\pm.0019}$	$.0053_{\pm.0005}$	$.0043_{\pm.0003}$	$.0167_{\pm.0010}$	$.0097_{\pm.0004}$	$.0075_{\pm.0002}$	$.0073_{\pm.0004}$	$.0062_{\pm.0002}$	$.0060_{\pm.0002}$
AlphaRec (BGE)	$.0380_{\pm.0059}$	**$.0241_{\pm.0019}$**	**$.0187_{\pm.0011}$**	$.0121_{\pm.0019}$	$.0083_{\pm.0006}$	**$.0072_{\pm.0003}$**	$.0166_{\pm.0010}$	$.0110_{\pm.0004}$	$.0079_{\pm.0002}$	$.0113_{\pm.0005}$	**$.0126_{\pm.0003}$**	**$.0119_{\pm.0002}$**
AlphaRec (Qwen)	$.0411_{\pm.0063}$	$.0213_{\pm.0018}$	$.0194_{\pm.0011}$	$.0127_{\pm.0019}$	$.0092_{\pm.0006}$	$.0080_{\pm.0004}$	$.0185_{\pm.0010}$	$.0120_{\pm.0004}$	$.0089_{\pm.0002}$	$.0135_{\pm.0006}$	$.0129_{\pm.0003}$	$.0128_{\pm.0002}$
ScaL^{3}AE (BGE)	**$.0443_{\pm.0065}$**	$.0210_{\pm.0017}$	$.0175_{\pm.0010}$	**$.0171_{\pm.0023}$**	**$.0106_{\pm.0007}$**	**$.0072_{\pm.0004}$**	**$.0360_{\pm.0015}$**	**$.0211_{\pm.0006}$**	**$.0127_{\pm.0003}$**	**$.0116_{\pm.0006}$**	$.0095_{\pm.0003}$	$.0076_{\pm.0002}$
ScaL^{3}AE (Qwen)	$.0509_{\pm.0072}$	$.0212_{\pm.0017}$	$.0180_{\pm.0010}$	$.0161_{\pm.0024}$	$.0099_{\pm.0007}$	$.0073_{\pm.0004}$	$.0417_{\pm.0016}$	$.0252_{\pm.0006}$	$.0181_{\pm.0004}$	$.0120_{\pm.0006}$	$.0086_{\pm.0003}$	$.0075_{\pm.0002}$

Table 3: NDCG@20 across 4 Amazon product domains, under the same conventions as Table 2.

	Games			Toys			Books			Electronics		
Model	10	5	3	10	5	3	10	5	3	10	5	3
EASE	$.0213_{\pm.0033}$	$.0095_{\pm.0010}$	$.0072_{\pm.0005}$	$.0077_{\pm.0012}$	—	—	$.0174_{\pm.0007}$	—	—	$.0050_{\pm.0002}$	—	—
SANSA	$.0220_{\pm.0035}$	$.0074_{\pm.0008}$	$.0046_{\pm.0004}$	$.0075_{\pm.0012}$	$.0037_{\pm.0003}$	$.0022_{\pm.0002}$	$.0187_{\pm.0008}$	$.0101_{\pm.0003}$	$.0065_{\pm.0002}$	$.0046_{\pm.0002}$	$.0028_{\pm.0001}$	$.0020_{\pm.0001}$
ELSA	$.0230_{\pm.0038}$	$.0090_{\pm.0009}$	$.0068_{\pm.0005}$	$.0074_{\pm.0012}$	$.0038_{\pm.0003}$	$.0031_{\pm.0002}$	$.0171_{\pm.0008}$	$.0085_{\pm.0003}$	$.0058_{\pm.0002}$	$.0056_{\pm.0003}$	$.0051_{\pm.0001}$	$.0044_{\pm.0001}$
L^3AE (BGE)	$.0225_{\pm.0035}$	$.0105_{\pm.0010}$	$.0077_{\pm.0005}$	$.0093_{\pm.0013}$	—	—	$.0168_{\pm.0007}$	—	—	$.0052_{\pm.0002}$	—	—
L^3AE (Qwen)	$.0230_{\pm.0036}$	$.0106_{\pm.0010}$	$.0079_{\pm.0005}$	$.0103_{\pm.0014}$	—	—	$.0169_{\pm.0007}$	—	—	$.0053_{\pm.0002}$	—	—
SGFCF	**$.0242_{\pm.0037}$**	$.0109_{\pm.0010}$	$.0067_{\pm.0005}$	$.0083_{\pm.0012}$	$.0032_{\pm.0003}$	$.0023_{\pm.0001}$	$.0114_{\pm.0006}$	$.0038_{\pm.0002}$	$.0016_{\pm.0001}$	$.0045_{\pm.0002}$	$.0035_{\pm.0001}$	$.0031_{\pm.0001}$
LightGCN	$.0211_{\pm.0033}$	$.0103_{\pm.0009}$	$.0082_{\pm.0005}$	$.0054_{\pm.0008}$	$.0038_{\pm.0003}$	$.0030_{\pm.0002}$	$.0087_{\pm.0005}$	$.0050_{\pm.0002}$	$.0035_{\pm.0001}$	$.0037_{\pm.0002}$	$.0035_{\pm.0001}$	$.0035_{\pm.0001}$
VBPR (BGE)	$.0193_{\pm.0033}$	$.0072_{\pm.0008}$	$.0054_{\pm.0004}$	$.0063_{\pm.0010}$	$.0030_{\pm.0003}$	$.0020_{\pm.0001}$	$.0104_{\pm.0006}$	$.0052_{\pm.0002}$	$.0038_{\pm.0001}$	$.0037_{\pm.0002}$	$.0030_{\pm.0001}$	$.0028_{\pm.0001}$
VBPR (Qwen)	$.0189_{\pm.0032}$	$.0081_{\pm.0009}$	$.0053_{\pm.0004}$	$.0065_{\pm.0011}$	$.0028_{\pm.0003}$	$.0021_{\pm.0001}$	$.0101_{\pm.0005}$	$.0053_{\pm.0002}$	$.0038_{\pm.0001}$	$.0037_{\pm.0002}$	$.0029_{\pm.0001}$	$.0026_{\pm.0001}$
AlphaRec (BGE)	$.0234_{\pm.0038}$	**$.0128_{\pm.0011}$**	**$.0091_{\pm.0006}$**	$.0084_{\pm.0012}$	$.0044_{\pm.0003}$	$.0035_{\pm.0002}$	$.0104_{\pm.0006}$	$.0058_{\pm.0002}$	$.0039_{\pm.0001}$	$.0057_{\pm.0003}$	**$.0058_{\pm.0001}$**	**$.0052_{\pm.0001}$**
AlphaRec (Qwen)	$.0235_{\pm.0034}$	$.0117_{\pm.0010}$	$.0093_{\pm.0006}$	$.0084_{\pm.0011}$	$.0049_{\pm.0003}$	$.0040_{\pm.0002}$	$.0114_{\pm.0006}$	$.0064_{\pm.0002}$	$.0043_{\pm.0001}$	$.0066_{\pm.0003}$	$.0059_{\pm.0001}$	$.0057_{\pm.0001}$
ScaL^{3}AE (BGE)	$.0239_{\pm.0032}$	$.0114_{\pm.0009}$	$.0081_{\pm.0005}$	**$.0113_{\pm.0014}$**	**$.0054_{\pm.0004}$**	**$.0037_{\pm.0002}$**	**$.0209_{\pm.0008}$**	**$.0118_{\pm.0003}$**	**$.0073_{\pm.0002}$**	**$.0058_{\pm.0003}$**	$.0046_{\pm.0001}$	$.0035_{\pm.0001}$
ScaL^{3}AE (Qwen)	$.0267_{\pm.0035}$	$.0115_{\pm.0009}$	$.0083_{\pm.0005}$	$.0108_{\pm.0015}$	$.0053_{\pm.0004}$	$.0037_{\pm.0002}$	$.0247_{\pm.0009}$	$.0127_{\pm.0003}$	$.0093_{\pm.0002}$	$.0060_{\pm.0003}$	$.0042_{\pm.0001}$	$.0035_{\pm.0001}$

Table 4: Head/tail Recall@20 on the three largest catalogs.

	Toys-3		Books-3		Electronics-3	
Method	Head	Tail	Head	Tail	Head	Tail
SANSA	.0040	.0036	.0176	.0079	.0052	.0031
ELSA	.0097	.0037	.0205	.0063	.0145	.0051
AlphaRec (BGE)	.0150	.0022	.0207	.0023	.0203	.0046
AlphaRec (Qwen)	.0166	.0025	.0225	.0029	.0215	.0053
ScaL^{3}AE (BGE)	.0072	.0073	.0191	.0099	.0086	.0067
ScaL^{3}AE (Qwen)	.0063	.0079	.0183	.0180	.0077	.0074

three largest catalogs (Table 4), whereas AlphaRec's overall score stems from a sharp head bias (up to ≈9× skew on Books-3) and both ELSA and SANSA from a milder collaborative imbalance toward head items. AlphaRec's contrastive loss captures popular-item signal well, while ScaL^{3}AE's semantic low-rank term fits cold items better — consistent with the k-core finding in §4.4. On Books and Toys the tail dominates, reversing the ranking — ScaL^{3}AE offers a strong semantic quality–scalability trade-off in colder regimes rather than the best on every column.

4.4 Cold-Item Analysis (RQ3)

A key design goal of ScaL^{3}AE is to bring semantic enrichment to large catalogs that contain many cold items. To test this, we vary the k-core threshold within each domain: lowering k-core from 10 to 3 grows the catalog, lowers density, and raises the fraction of cold items, while keeping the domain, embedding model, and split protocol fixed.

Figure 2 plots the relative Recall@20 lift of ScaL^{3}AE over SANSA at 10/5/3-core across all four domains. In three of four the lift grows monotonically as the catalog gets colder, a pattern consistent with semantic enrichment helping most where collaborative signals are weakest. Books under BGE is the only non-monotonic case, but switching to Qwen restores monotonicity — tracking a spectrum collapse of $\mathbf{F}\mathbf{F}^\top$ (top eigenvalue 58% under BGE vs 40% under Qwen) toward a single dominant direction — so the anomaly is embedding-specific. Note that varying k-core also shifts catalog composition, so this reads as a stress test across colder regimes rather than an isolated ablation of item coldness.

4.5 Residual Correction Ablation (RQ4)

Eq. (10) combines three terms: the SANSA prediction, the residual correction $(\mathbf{x} - \mathbf{x}\,\mathbf{B}_{\text{SANSA}}) \odot \mathbf{w}$, and the semantic low-rank term. The semantic term's contribution is already captured by ScaL^{3}AE vs.

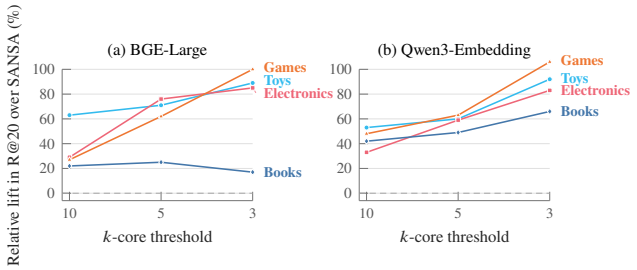


Figure 2: Relative Recall@20 lift of ScaL^{3}AE over SANSA as the k-core threshold decreases.

SANSA (§4.3); here we isolate the residual correction by comparing the full model against the $\mathbf{w} = \mathbf{0}$ variant.

Table 5: Residual correction ablation on ScaL^{3}AE. $\Delta = (R_{\text{full}} - R_{\text{no-res}})/R_{\text{no-res}}$ (positive = helps).

	Δ all	Δ head	Δ tail
BGE-Large			
Mean	−3.7%	+11.4%	−10.3%
Median	−8.0%	+11.2%	−13.0%
Range	[−16.3, +11.8]	[−2.7, +26.6]	[−26.8, +4.5]
Qwen3-Embedding			
Mean	−0.6%	+14.1%	−6.5%
Median	−0.4%	+14.9%	−7.0%
Range	[−10.8, +12.1]	[+3.9, +23.9]	[−17.6, +5.4]

The residual term acts as a head/tail calibration knob (Table 5): it lifts head recall but hurts the tail, with a small net penalty overall. We keep $\mathbf{w}\neq\mathbf{0}$ as the default, following L^{3}AE's zero-diagonal closed form; tail-dominated catalogs may prefer $\mathbf{w}=\mathbf{0}$.

5 Conclusion

ScaL^{3}AE pairs Woodbury factorization with SANSA's sparse inverse, cutting memory from $O(n^2)$ to $O(\rho n^2 + dn)$ while preserving L^{3}AE's algebraic structure; it outperforms EASE and SANSA on 4 Amazon domains with gains that grow in colder regimes.

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